• Properties of continuous functions

Using Limit Laws, we can prove that algebraic combinations of continuous functions are continuous everywhere they are defined.

Theorem

If f, g are continuous at $x = a$, then so are the following:

1. *Sums, differences: $f \pm g$*
2. *Constant multiple: $k \cdot f$, for any number k*
3. *Products: $f \cdot g$*
4. *Quotients: $\frac{f}{g}$, provided that $g(a) \neq 0$*
5. *Powers: f^n, where n is a positive integer*
6. *Roots: $\sqrt[n]{f}$, provided that it is defined on an open interval containing a, where n is a positive integer.*

• Continuity on an interval

So far we have only been discussing continuity at a single point, while in general we want to describe the continuity behavior of a function throughout its entire domain

Definition

- A function f is continuous on an interval if it is continuous at every number in the interval.
- If f is defined only on one side of an endpoint of the interval, we understand continuous at the endpoint to mean continuous from the right or continuous from the left.

Note that by interval we are including all of the following possibilities:

$$(a,b),\ (a,b],\ [a,b),\ [a,b],\ (-\infty,b),\ (-\infty,b],\ (a,\infty),\ [a,\infty),\ (-\infty,\infty).$$

Example

The function $f(x) = 2 - \sqrt{1 - x^2}$ is continuous on the interval $[-1, 1]$.

Solution.

- If $-1 < a < 1$, then using the Limit Laws, we have

$$\begin{aligned} \lim_{x \to a} f(x) &= \lim_{x \to a}(2 - \sqrt{1 - x^2}) = 2 - \lim_{x \to a} \sqrt{1 - x^2} \\ &= 2 - \sqrt{\lim_{x \to a}(1 - x^2)} = 2 - \sqrt{1 - a^2} = f(a). \end{aligned}$$

Thus f is continuous at a.

- At the end points ± 1, similar calculations show that

$$\lim_{x \to -1^+} f(x) = 2 = f(-1) \qquad \text{and} \qquad \lim_{x \to 1^-} f(x) = 2 = f(1).$$

So f is continuous from the right at -1 as well as from the left at 1.

So conclusion: f is continuous on $[-1, 1]$.

• Continuity of important functions

It turns out that most of the familiar functions are continuous at every number in their domains.

- Polynomials: $P(x) = \sum_{i=0}^{n} a_i x^i$
- Power functions: $f(x) = x^a$
- Rational functions: $f(x) = \dfrac{P(x)}{Q(x)}$
- Algebraic functions
- Trigonometric functions: $\cos x$, $\sin x$, $\tan x$, $\cot x$, $\sec x$, $\csc x$
- Exponential functions: $f(x) = a^x$ $(a > 0, \neq 1)$
- Logarithmic functions: $f(x) = \log_a x$ $(a > 0, \neq 1)$

We have the following principle.

- The general principle. Any function constructed from continuous functions by doing additions, multiplications, and divisions, is itself going to be continuous at every point of its domain.

This principle shows how to build up complicated continuous functions from simple ones by doing additions, multiplications, and divisions.

- Another way of combining continuous functions f and g to get a new continuous function is to form the composite function $f \circ g$.

This fact is a consequence of the following result.

Theorem

If f is continuous at b and $\lim_{x \to a} g(x) = b$, then $\lim_{x \to a} f\big(g(x)\big) = f(b)$.

In other words,

$$\lim_{x \to a} f\big(g(x)\big) = f\left(\lim_{x \to a} g(x)\right).$$

This theorem says that a limit symbol $\lim_{x \to a}$ can be moved through a function symbol f if the function is continuous and the limit exists.

In other words, the order of these two symbols can be reversed.

• Continuity of composite functions

Applying the previous theorem, we can get the fact that a continuous function f of a continuous function g is a continuous function $f \circ g$.

More precisely, we have the following result about composition of continuous functions:

If g is continuous at a and f is continuous at $g(a)$, then the composite function $f \circ g$ given by $(f \circ g)(x) = f\big(g(x)\big)$ is continuous at a.

Example

We have

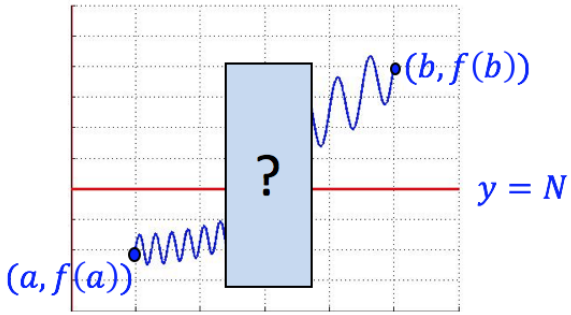
$$\lim_{x \to a} |g(x)| = \left| \lim_{x \to a} g(x) \right|,$$

provided that $\lim_{x \to a} g(x)$ exists.

Solution. We apply the theorem above with $f(x) = |x|$, which is continuous everywhere.

• Application of continuity

Let us consider a function $f(x)$ with domain $[a, b]$ whose graph is shown below. I have concealed part of the graph under a box.



Now I tell you that *"the graph of $f(x)$ intersect the line $y = N$ at no point"*.

Question. What can you conclude?

Answer. $f(x)$ cannot be continuous on the whole interval $[a, b]$.

• Intermediate Value Theorem

The observation we just made is actually an important theorem about an important property of continuous functions.

The Intermediate Value Theorem for a continuous function

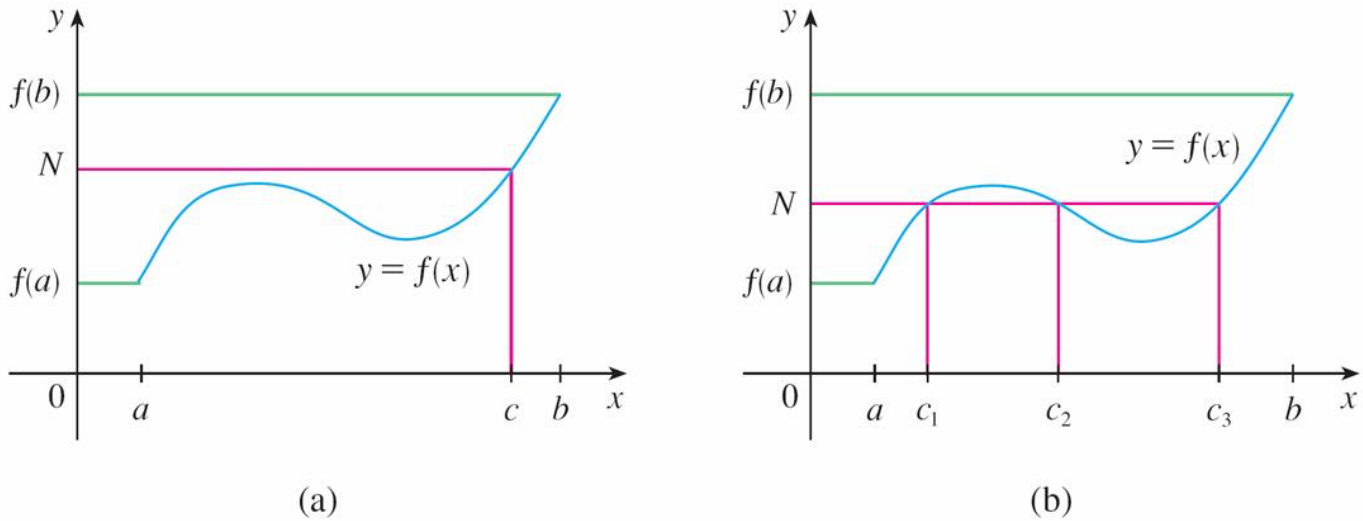
Let f be a function, and a, b, N real numbers with $a < b$. Suppose

- f is continuous on the closed interval $[a, b]$
- $f(a) \neq f(b)$
- N is in between $f(a)$ and $f(b)$.

Then there exists a number $c \in (a, b)$ such that

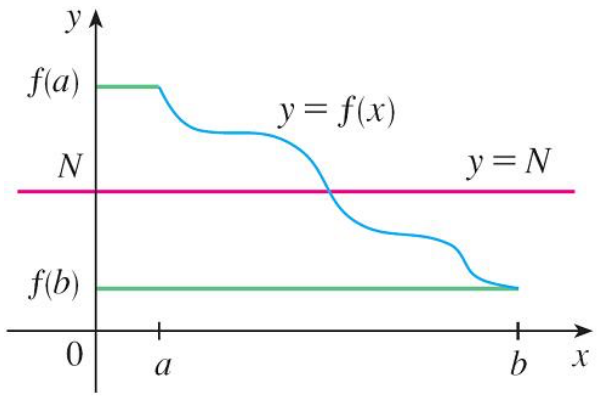
$$f(c) = N.$$

The IVT for continuous functions states that a continuous function takes on every intermediate value between the function values $f(a)$ and $f(b)$.



Note that the value N can be taken on once (as in part (a)) or more than once (as in part (b)).

If we think of a continuous function as a function whose graph has no hole or break, then it is easy to believe that the IVT is true.



In geometric terms it says that if any horizontal line $y = N$ is given between $y = f(a)$ and $y = f(b)$, then the graph of f cannot jump over the line. It must intersect $y = N$ somewhere.

Example

Show that the equation

$$4x^3 - 6x^2 + 3x - 2 = 0$$

has a root in between 1 and 2.

Solution. Let $f(x) = 4x^3 - 6x^2 + 3x - 2$. We have to show that there exists a number $c \in (1, 2)$ such that $f(c) = 0$.

Taking $a = 1$, $b = 2$, and $N = 0$ in the IVT, we have

$$f(1) = 4 - 6 + 3 - 2 = -1 < 0$$
$$f(2) = 32 - 24 + 6 - 2 = 12 > 0.$$

Thus $f(1) < 0 < f(2)$; that is, $N = 0$ is a number between $f(1)$ and $f(2)$.

Now f is continuous since it is a polynomial, so the IVT says that there is a number $c \in (1, 2)$ such that $f(c) = 0$.

In other words, the equation $4x^3 - 6x^2 + 3x - 2 = 0$ has at least one root c in the interval $(1, 2)$.

Comment. In fact, we can locate a root more precisely by using the IVT again. For instance, it is easy to check that

$$f(1.2) = -0.128 < 0 \quad \text{and} \quad f(1.3) = 0.548 > 0,$$

and so a root must be in between 1.2 and 1.3.

Applications of continuity in real life

The concept of continuity of functions has **various applications** in real life across different fields.

These are some examples showcasing the wide range of applications of continuity in various fields.

The concept's importance lies in providing a mathematical framework to describe and understand the smooth behavior of real-world phenomena, leading to practical applications and insights.

• **Engineering & Physics**: Continuity is crucial in engineering and physics to ensure smooth operations and avoid abrupt changes. For instance, in designing structures like bridges or buildings, engineers consider the continuity of stress and strain to ensure the stability and durability of the structure. In physics, the concept of continuity is employed to describe the flow of fluids, such as in fluid mechanics or electrical circuits, ensuring the smooth transfer of energy.

• **Computer Science**: Continuity plays a role in computer graphics and image processing, where smoothness is desired. For example, in rendering 3D graphics or creating animations, the concept of continuity is used to ensure that the images appear seamless and visually appealing. In computer simulations and modeling, continuity is employed to represent the smooth evolution of a system over time.

• **Environmental Sciences**: Continuity is significant in the study of environmental phenomena, such as weather forecasting, climate modeling, and pollution dispersion. Weather and climate models rely on the continuity of atmospheric variables to simulate and predict weather patterns accurately. Pollution dispersion models utilize continuity to track the spread of pollutants in the atmosphere or water bodies.

• **Medicine and Biology**: Continuity is relevant in medical and biological applications. In medical imaging, techniques like computed tomography (CT) scans or magnetic resonance imaging (MRI) rely on the continuity of body structures to construct accurate images. In biology, continuity is employed to describe biological processes, such as growth, cell division, or metabolic reactions, which often exhibit smooth and continuous changes.

• **Economics and Finance**: Continuity is often employed in economic and financial models to represent the smooth progression of variables. Economic theories assume that production, consumption, and investment functions are continuous to predict the behavior of markets and make informed decisions. In finance, continuity is used in options pricing models, portfolio optimization, and risk management.

END OF CHAPTER 1

CALCULUS I - AY2024-25

Chapter 2: Derivatives

Outline

1. Derivatives of Functions
2. Differentiation Rules
3. The Chain Rule
4. Implicit Differentiation
5. Related Rates
6. Linear Approximations

Summary

Derivative is one of the key ideas in calculus, which is used to study many problems, not only in mathematics, but also in say

- science: *determining the rate of flow of a liquid into a container, ...*
- economics: *finding the number of items a manufacturing company should produce to maximize its profits, ...*
- medicine: *calculating the amount of blood the heart pumps, ...*

The main topics of this chapter are

- The derivative as a function
- The relationship between continuity and differentiability
- Higher derivatives

• Derivatives and rates of change

Do you remember the main motivation for introducing limits?

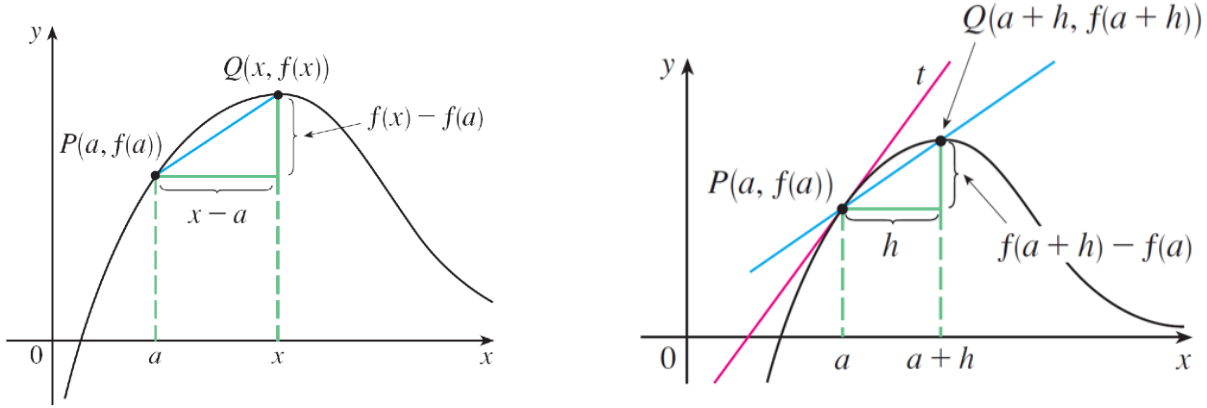
We discovered a need of limits when solving tangent problems.

Definition

- Let $D \subset \mathbb{R}$ and let $f\colon D \to \mathbb{R}$ be a function. We define the tangent (or tangent line) to the curve $y = f(x)$ (or graph of $f(x)$) at the point a to be the line going through the point $(a, f(a))$ with slope

$$m = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} \qquad \text{(if the limit exists).}$$

- If this limit does not exist then we say that the graph does not have a tangent at that point.



Instead of working with the position x of the second point (left figure), we could equally well work with the displacement h from a to that point (right figure). Then the slope of the tangent line becomes

$$m = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

• Limit and the velocity of an object

Example

Suppose that a ball is dropped from the upper observation deck of the “CN Tower” (Canadian National Tower, Toronto) 450 m above the ground. Find the velocity of the ball after say 5 seconds.

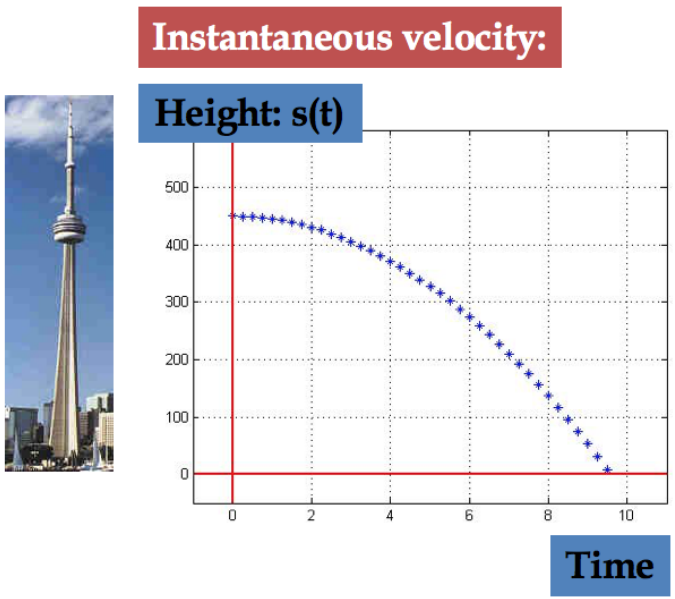




Definition

Suppose that a particle's trajectory (in one dimension) is described by a function $s : \mathbb{R} \to \mathbb{R}$. The instantaneous velocity of the particle at time t is defined by

$$\lim_{\tau \to 0} \frac{s(t+\tau) - s(t)}{\tau}.$$



• Derivative of a function

Since this type of limit $\lim_{h\to 0} \dfrac{f(a+h)-f(a)}{h}$ actually occurs widely and arises whenever we calculate a rate of change in any of the sciences or engineering, it is given a special name and notation.

Definition. The derivative of a function f at a point a, denoted by $f'(a)$, is

$$f'(a) = \lim_{h\to 0} \frac{f(a+h)-f(a)}{h},$$

provided this limit exists.

An equivalent way of stating the definition of the derivative, as we saw in finding tangent lines, is

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

We can now say that the tangent line to $y = f(x)$ at $(a, f(a))$ is the line through $(a, f(a))$ whose slope is equal to $f'(a)$, the derivative of f at a.

Suppose y is a quantity that depends on another quantity x, i.e. y is a function of x and we write $y = f(x)$. Then we can have another interpretation of $f'(a)$: the derivative $f'(a)$ is the instantaneous rate of change of $y = f(x)$ with respect to x when $x = a$.

• The derivative as a function

We have considered the derivative of a function f at a fixed number a

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

Now we investigate the derivative as a function derived from f by considering the limit at each point x in the domain of f.

Definition

The derivative of a function $f(x)$ (w.r.t. the variable x) is defined by the following rule:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}, \quad \text{if this limit exists at } x.$$

Note. In general, the domain of $f'(x)$, which is the set $\{x : f'(x) \text{ exists}\}$, may be *smaller* than the domain of f.

Other notations for the derivative. Some common alternative notations for the derivative of a function $y = f(x)$ are as follows:

$$f'(x) = y' = \frac{dy}{dx} = \frac{df}{dx} = \frac{d}{dx} f(x).$$

Here the symbol $\frac{d}{dx}$ are called differentiation operators because they indicate the operation of differentiation, which is the process of calculating a derivative.

Note. The symbol $\frac{dy}{dx}$, which was introduced by Leibniz, should not be regarded as a ratio (for the time being); it is simply a synonym for $f'(x)$. We can rewrite the definition of derivative in Leibniz notation in the form

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{x_2 \to x_1} \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

The notation

$$\left.\frac{dy}{dx}\right|_{x=a}$$

is a synonym for $f'(a)$. The vertical bar means *"evaluate at"*.



Gottfried Wilhelm (von) Leibniz (sometimes spelled Leibnitz), 1646 - 1716, was a prominent German mathematician and philosopher.

- His ideas of differential and integral calculus (independently of Isaac Newton's developments) were most notable achievement.
- He also refined the binary number system, which is the foundation of all digital computers.

Definition

- A function $f(x)$ is differentiable at a point a if $f'(a)$ exists. It is differentiable on an open interval (finite or infinite) (a, b), or (a, ∞), or $(-\infty, a)$, or $(-\infty, \infty)$, if it is differentiable at every point of the interval.
- It is differentiable on a closed interval $[a, b]$ if it is differentiable on the interior interval (a, b) and if the one-side limits at a

$$\lim_{h \to 0^+} \frac{f(a+h) - f(a)}{h}, \qquad \lim_{h \to 0^-} \frac{f(a+h) - f(a)}{h}$$

exist.

Both continuity and differentiability are desirable properties for a function to have. The following result shows how these properties are related.

Theorem

If $f(x)$ is differentiable at a, then f is continuous at a.

Note. The inverse implication of the theorem above is false. That is, there are functions that are continuous but not differentiable.

Example

Determine the derivative of $f(x) = |x|$. State the domain of f'

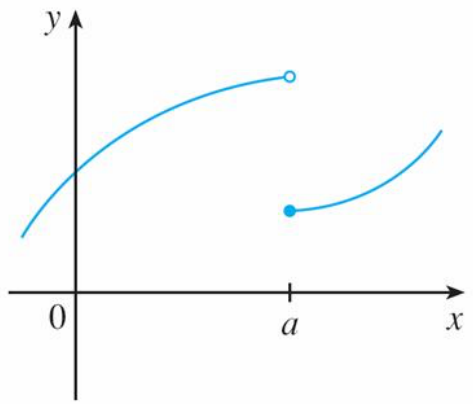
Solution. We have $f'(x) = \begin{cases} -1, & \text{if } x < 0 \\ 1, & \text{if } x > 0 \end{cases}$ is not differentiable at $x = 0$.

Question. When does a function not have a derivative at a point?

Answer. There are mainly 3 ways that a function can fail to be differentiable at some point a.

- Possibility 1. f is not continuous at a.

It is clear, as f cannot be differentiable at a if it is not continuous at a.



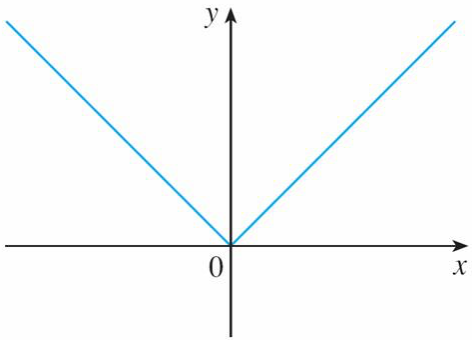
- Possibility 2. f has a corner at a.

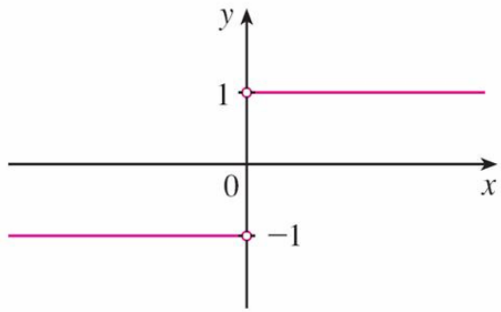
This means that

$$\lim_{h\to 0^-} \frac{f(x+a)-f(a)}{h} \qquad \text{and} \qquad \lim_{h\to 0^+} \frac{f(x+a)-f(a)}{h}$$

both exist, but they are different.

Look back at the previous example $f(x) = |x|$ at $x = 0$



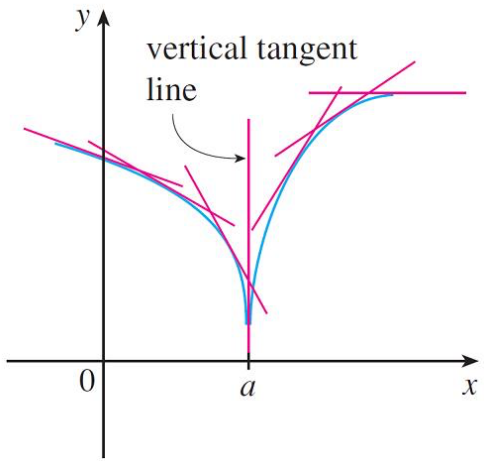


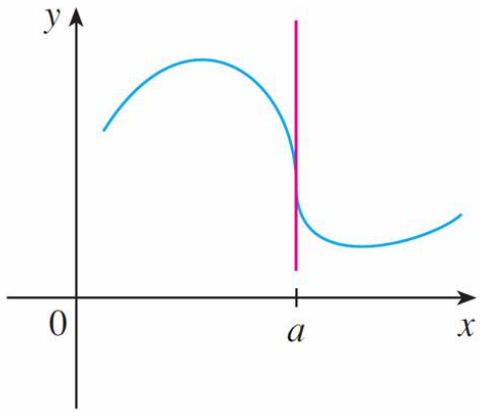
- Possibility 3. f has a vertical tangent at a.

This means that

$$\lim_{h\to 0} \frac{f(a+h)-f(a)}{h} = \infty \quad \text{or} \quad -\infty,$$

indicating that the tangent lines become steeper and steeper as $x \to a$.

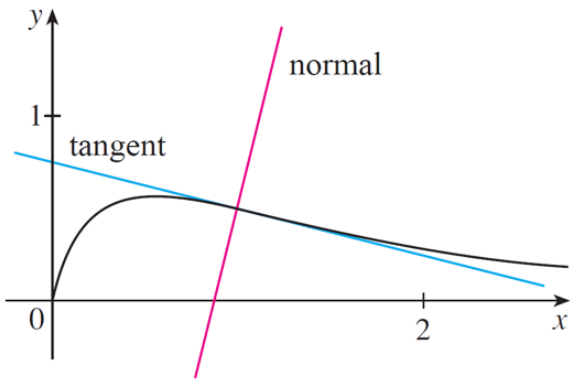




Definition

The normal line to a curve $\mathcal{C}$ at a point P is the line through P that is perpendicular to the tangent line at P.

The curve and its tangent and normal lines are graphed in the figure below.



• Higher derivatives

- We know that if f is a differentiable function, then its derivative f' is also a function. So f' may have a derivative of its own, denoted by $(f')' = f''$.

This new function f'' is called the second derivative of f (because it is the derivative of the derivative of f).

Using Leibniz notation, we write the second derivative of $y = f(x)$ as

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2}.$$

- The third derivative f''' is the derivative of the second derivative: $f''' = (f'')'$. Likewise, it can be interpreted as the slope of the curve $y = f''(x)$ or as the rate of change of $f''(x)$.

If $y = f(x)$, then alternative notations for the third derivative are

$$y''' = f'''(x) = \frac{d}{dx}\left(\frac{d^2y}{dx^2}\right) = \frac{d^3y}{dx^3}.$$

The differentiation process can be continued.

- In general, the n-th derivative of f, denoted by $f^{(n)}$, is obtained from f by differentiating n times.

If $y = f(x)$, we write

$$y^{(n)} = f^{(n)}(x) = \frac{d}{dx}\left(\frac{d^{n-1}y}{dx^{n-1}}\right) = \frac{d^ny}{dx^n}.$$

• Acceleration

The second derivative has a special meaning. It can be interpreted as a rate of change of a rate of change.

The most familiar example of this is acceleration, which we define as follows.

If $s = s(t)$ is the position function of an object that moves in a straight line, then (we know that) its first derivative represents the **velocity** $v(t)$ of the object as a function of time:

$$v(t) = s'(t) = \frac{ds}{dt}.$$

The instantaneous rate of change of velocity with respect to time is called the **acceleration** $a(t)$ of the object.

Thus the acceleration function is the derivative of the velocity function and hence it is the second derivative of the position function:

$$a(t) = v'(t) = s''(t),$$

or in Leibniz notation

$$a(t) = \frac{dv}{dt} = \frac{d^2s}{dt^2}.$$

• Speed

Let us consider a particle that is moving in a straight line. Denote by $s = f(t)$ the position function of at a time t.

We already defined

- the *velocity* $v(t) = s' = \dfrac{ds}{dt}$ (the rate of change of position w.r.t. time), and
- the *acceleration* $a(t) = v'(t) = s''(t)$ (the change of velocity w.r.t. time).

We now define also the **speed** which is the **absolute value** of velocity $|v(t)|$.

Exercise

The position of a particle is given by $s = f(t) = t^3 - 6t^2 + 9t$, where t is measured in seconds and s in meters.

(1) Find the velocity at time t.

(2) When is the particle at rest?

(3) When is the particle moving forward (that is, in the positive direction)?

(4) Draw a diagram to represent the motion of the particle.

(5) Find the total distance traveled by the particle during the first 5 sec.

(6) Find the acceleration at time t and after 4 sec.

(7) Graph the position, velocity, and acceleration functions.

(8) When is the particle speeding up? When is it slowing down?

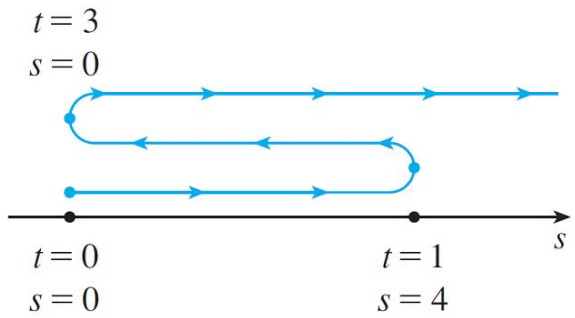
Answer.

(1) $v(t) = 3t^2 - 12t + 9$.

(2) $t = 1$, $t = 3$ (sec).

(3) $t > 3$ (sec), or $t < 1$ (sec).

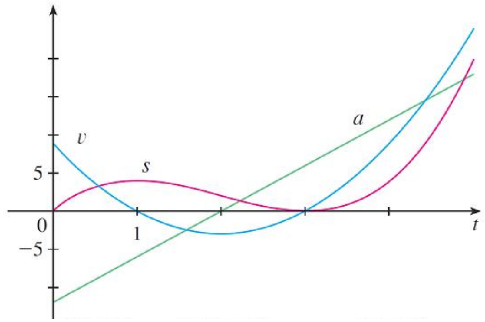
(4)



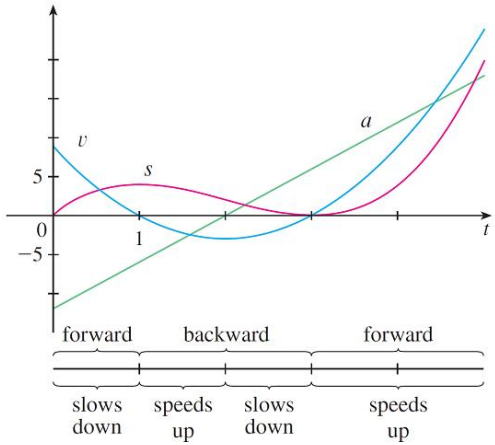
(5) 28 (m).

(6) 12 (m/sec^2).

(7)


v
s
a
5
0
−5
1
t

(8)


v
s
a
5
0
−5
1
t
forward
backward
forward
slows down
speeds up
slows down
speeds up